Sèrie 2

Exercici 1

Q1 a

Q2 c

Q3 b

Q4 a

Q5 c

Exercici 2

a)

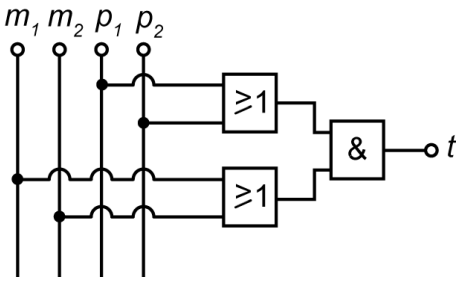
m_1	m_2	p_1	p_2	t
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

b)

$$t = (m_1 + m_2 + p_1 + p_2)(m_1 + m_2 + p_1 + \bar{p}_2)(m_1 + m_2 + \bar{p}_1 + p_2)$$
$$(m_1 + m_2 + \bar{p}_1 + \bar{p}_2)(m_1 + \bar{m}_2 + p_1 + p_2)$$
$$(\bar{m}_1 + m_2 + p_1 + p_2)(\bar{m}_1 + \bar{m}_2 + p_1 + p_2)$$

simplificant: $t = (m_1 + m_2)(p_1 + p_2)$

c)



Exercici 3

a)

$$\Gamma_{mot} = \frac{P_{mot}}{\omega_{mot}} = \frac{P_{mot}}{n_{mot}\frac{2\pi}{60}} = 3{,}712 \text{ Nm}$$

b)

$$\Gamma_{red} = \frac{P_{red}}{\omega_{red}} = \frac{\eta P_{mot}}{\omega_{red}} = \frac{\eta \Gamma_{mot}}{\tau} = 51{,}72 \text{ Nm}$$

c)

$$n_d = n_{mot}\tau = 97{,}49 \text{ min}^{-1}$$

d)

$$n_{\text{bombo}} = n_d \frac{d}{D} = 12{,}28 \text{ min}^{-1}$$

e)

$$\Gamma_{\text{res}} = \frac{\eta P_{\text{mot}}}{\omega_{\text{bombo}}} = \frac{\eta P_{\text{mot}}}{n_{\text{bombo}} \frac{2\pi}{60}} = 410{,}4 \text{ Nm}$$

Exercici 4

a)

$$E_{\text{cons}} = P_{\text{inst}} c\,t = 98550 \text{ kWh anuals} = 354780 \text{ MJ anuals}$$

b)

$$P_{\text{fotovolt}} = r\,P_{\text{cons}} c = 3{,}375 \text{ kW}$$

c)

$$\eta_{\text{placa}} = \frac{\frac{P_{\text{placa}}}{A}}{I_{\text{rad}}} = 13{,}38\%$$

d)

$n_{\text{p}} = \frac{P_{\text{foto}}}{P_{\text{placa}}} = 17{,}4$ plaques. Calen 18 plaques

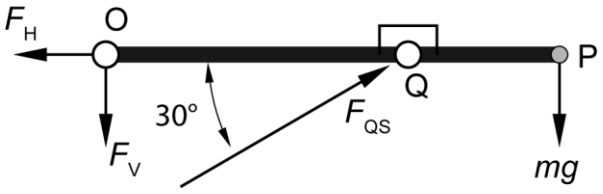
e)

Si s'utilitzen 18 plaques $r' = 15{,}52\%$

$\Delta m = r' E_{\text{cons}} FE = 3{,}686$ tones de CO_2

Exercici 5

a)



b)

$$\sum M(O) = 0;$$

$$F_{QS}\sin(30)\,4L = mg\,6L \;\rightarrow\; F_{QS} = 3mg = 2354\,\text{N}$$

La barra treballa a compressió

c)

$$\sum F_{\text{horitzontals}} = 0;\;\; F_H = F_{QS}\cos(30) = 3mg\sqrt{3}/2 = 2038\,\text{N}$$

$$\sum F_{\text{verticals}} = 0;\;\; F_V + mg = F_{QS}\sin(30) \;\rightarrow\; F_V = 392{,}3\,\text{N}$$

Exercici 6

a)

$$\eta_{\text{alt}} = \frac{P_{\text{electr}}}{P_{\text{mot}}} = 73{,}76\%$$

b)

$$c_{\text{gasoil}} = \frac{\rho_{gasoil}\,V}{t} = 915{,}4\ \text{g/h}$$

c)

$$\eta = \frac{P_{mot}}{P_{cons}} = \frac{P_{mot}}{c_{gasoil}\ p_c} = 65,46\%$$

d)

$$P_{diss} = P_{cons} - P_{electr} = 5,891\ \text{kW}$$